The control variable accesses each item of the sequence on each iteration until it reaches the last item in the sequence.

Example 6.8 (a)

```
for x in "Hello World":
    print(x, end=' ')
```

Output:

```
H e l l o  W o r l d
```

Example 6.8 (b)

```
for x in (1,2,3,4,5):
    print("Hello World")
```

Output:

```
Hello World
Hello World
Hello World
Hello World
Hello World
```

In the above example 6.8(a), we have created a string "Hello World", as the sequence.

Initially, the value of x is set to the first element of the string, (i.e. 'H'), so the print statement inside the loop is executed. Then, the control variable x is updated with the next element of the string and the print statement is executed again. In this way the loop runs until the last element of the string is accessed.

In the same way, example 6.8(b) prints the string "Hello World", five times, until the control variable x reaches last element of the given sequence.

Instead of creating sequence of values manually, we can use range().

The range() is a built-in function, to generate series of values between two numeric intervals.

The syntax of range() is as follows:

range (start,stop,[step])

Where,

start – refers to the initial value

stop – refers to the final value

step – refers to increment value, this is optional part.

Example 6.8(c): Examples for range()

```
range (1,30,1)   will start the range of values from 1 and end at 29
range (2,30,2)   will start the range of values from 2 and end at 28
range (30,3,-3)  - will start the range of values from 30 and end at 6
range (20)       will consider this value 20 as the end value(or upper limit) and starts the
                 range count from 0 to 19 (remember always range() will work till stop -1
                 value only)
```


12th Computer Science_EM Chapter 6.indd 77 26-12-2022 16:44:46